

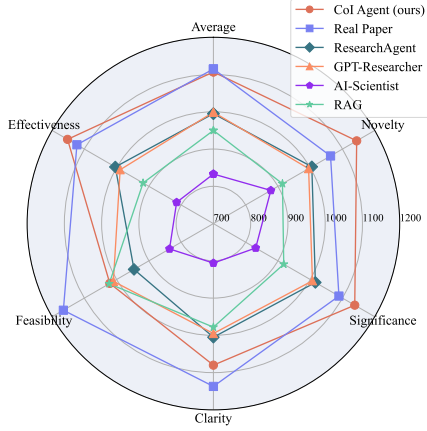


Figure 3: Evaluation results of idea generation with LLM as a judge.

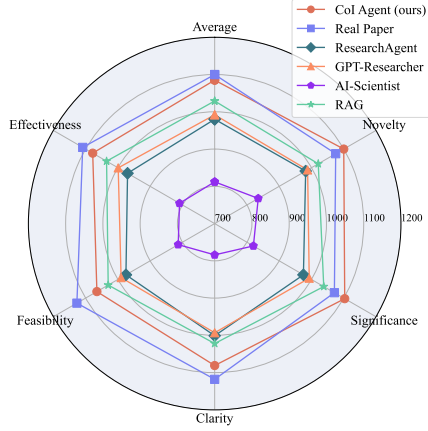


Figure 4: Evaluation results of idea generation with human as judges.

between automatic methods and Real Paper is expected, as Real Paper ideas undergo extensive experimental validation. Additionally, AI-Scientist’s performance is especially low, likely due to its original design, which focuses on generating full papers from executable code. When given only a research topic, its simplistic idea generation framework limits its ability to produce novel and feasible ideas.

Table 1: Agreement between the human and GPT-4o judges in all evaluated dimensions.

	Novelty	Significance	Clarity	Feasibility	Effectiveness	Average
Agreement	66.5%	71.0%	76.3%	70.2%	71.0%	70.8%

Human-Model Agreements of Idea Arena. To assess the reliability of our model-based evaluation within Idea Arena, we analyze the agreements between the preferences of the human judges and the LLM judges. We follow Zheng et al. (2024) to compute the agreement, which is defined as the probability that two judges agree on the winner of one specific arena match. Figure 5 shows the pairwise agreement between humans and several state-of-the-art LLMs, including GPT-4o, Gemini-1.5-Pro-Exp-0827⁷, and Claude-3.5-Sonnet⁸. We observe an average agreement of 70.8% between GPT-4o and humans. This finding indicates a strong alignment between human-based and model-based evaluations, thereby highlighting the robustness of Idea Arena in evaluating the quality of generated research ideas. Moreover, GPT-4o demonstrates the highest level of agreement with humans among all tested LLMs. Therefore, we will utilize GPT-4o as the LLM judge for subsequent analytical experiments. Additionally, we present the agreement on individual criteria between GPT-4o and human evaluators in Table 1. The results indicate a consistently high level of agreement across all assessed criteria.

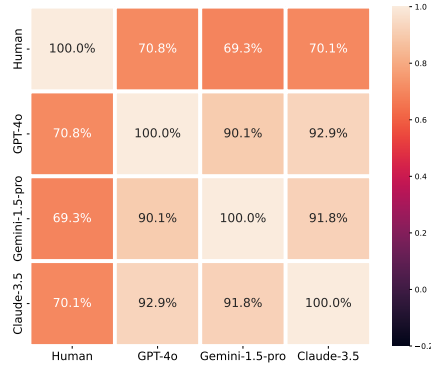


Figure 5: Agreements between human and LLM judges.

4.2 ABLATION STUDIES FOR IDEA GENERATION

We conduct an ablation study to assess the contributions of each component of the CoI Agent to idea generation quality. The following variants are examined: 1) – *CoI*: Excludes the CoI construction stage, directly using all retrieved literature without progressive relation mining. 2) – *Future Trend*:

⁷<https://ai.google.dev/gemini-api/docs/models/experimental-models>

⁸<https://www.anthropic.com/news/claude-3-5-sonnet>

Table 2: Ablation study on the design of CoI agent. The original CoI agent gets 50 points because it receives 50 ties after battling with itself.

	Novelty	Significance	Clarity	Feasibility	Effectiveness	Average
CoI Agent	50	50	50	50	50	50
– CoI	41	39	44	49	39	42.4
– Future Trend	40	43	51	53	44	46.2
– Entities	46	49	42	47	43	45.4

Omits the Future Trend Prediction module, prompting the LLM to consolidate ideas directly based on the provided input information. 3) – *Entities*: Skips inputting entity definitions during idea generation. To ensure fair comparison, each variant is scored against the full CoI Agent, with 2/1/0 points for win/tie/lose in 50 matches, for a maximum of 100 points.

Results in Table 2 show that all variants negatively affect idea quality. Excluding the CoI construction stage has the most significant impact, emphasizing the importance of organizing literature based on progressive relationships to enhance the LLM’s understanding of trends. Removing the Future Trend Prediction reduces novelty, as the LLM lacks insight into potential forward-thinking ideas. Although slight improvements in clarity and feasibility are observed, these are not substantial, likely due to evaluation variability. Finally, omitting entity information reduces clarity and effectiveness, as the LLM generates more abstract ideas without grounding in specific concepts. This highlights the value of entity information in enhancing the clarity and practical relevance of ideas.

4.3 CASE STUDY

We present an intriguing case study in Table 3 with the same topic of our paper – generating novel research ideas using LLMs. Given the input topic, our CoI agent first constructs the chain of ideas, extending I_0 (Baek et al., 2024) in both forward and backward directions. Then the agent analyzes current research trends for any two adjacent ideas. For instance, it identifies that the core development from I_{-1} to I_0 is the generation of ideas rather than hypotheses. After digesting the existing trends, the CoI agent realizes that LLMs have great potential in idea generation but are limited in novelty and diversity. Therefore, it proposes an evolutionary algorithm, which specifically models the variations between parents and children, as a possible future trend for novel and diverse idea generation. Finally, the agent consolidates its final idea by drawing on future trends and with practical implementations, such as crossover and mutation, to ensure effective realization. Therefore, the generated idea is viable and novel, deserving further exploration in our future work.

4.4 EXPERIMENT DESIGN

As a byproduct of idea generation, we also require these baselines to develop potential experiment designs for realizing their proposed ideas. Table 4 presents the arena-style results for experiment designs for both model-based and human-based evaluations. Our CoI Agent demonstrates superior performance across all evaluated criteria in two evaluation settings, achieving the highest scores among all automated methods. Notably, it surpasses RAG, the second-best automated method, by 70 ELO points in human evaluation. Furthermore, there is a high degree of model-human agreement in the experimental designs. Despite the clarity and reasonable technical details of the experiment designs produced by the CoI Agent in support of the proposed ideas, they tend to be less feasible compared to those designs in the existing literature. This phenomenon is also observed during the idea generation phase. Consequently, feasibility represents a significant bottleneck in automatic idea generation, highlighting the need for future research to address this challenge.

Table 4: Results of experiment design including both model and human evaluations, as well as their agreements. Tech. refers to the Technical Quality criterion.

		Feasibility	Tech.	Clarity	Average
Model Evaluation	Real Paper	1100	1122	1090	1103
	CoI Agent (ours)	1029	1096	1043	1056
	RAG	1022	970	1016	1003
	ResearchAgent	960	1020	980	987
	GPT-Researcher	1001	965	992	986
	AI-Scientist	888	827	879	865
Human Evaluation	Real Paper	1138	1111	1111	1120
	CoI Agent (ours)	1092	1123	1121	1112
	RAG	1035	1041	1048	1042
	GPT-Researcher	988	977	971	978
	ResearchAgent	939	959	964	954
	AI-Scientist	809	788	785	794
	Agreement	70.7%	75.9%	72.1%	73.0%

Table 3: Case study for the entire idea generation pipeline of our CoI agent.

Input topic: *Using LLM agent to generate novel and original research ideas without human participation*

Chain of ideas:

- I_{-3} (Kim et al., 2021): It addresses the challenge of discovering new materials through molecular generation. It introduces GCT, a Transformer with a variational autoencoder, to generate SMILES strings . . .
 - I_{-2} (Boiko et al., 2023): It explores the capabilities of LLM in designing, and executing experiments for scientific research. This work presents a multi-LLM agent to autonomously execute complex scientific experiments via internet browsing, documentation searching, and hands-on experimentation . . .
 - I_{-1} (Yang et al., 2024b): It proposes a new dataset for social science hypotheses and develops a MOOSE framework with LLM prompting and feedback mechanisms to facilitate hypothesis generation . . .
 - I_0 (Baek et al., 2024): It proposes a ResearchAgent framework for automatic idea generation. ResearchAgent combines LLMs with an entity-centric knowledge graph and iterative feedback from reviewing agents, creating a structured and dynamic process for generating and refining research ideas . . .
 - I_1 (Si et al., 2024): The paper explores the capabilities of LLMs in generating novel research ideas and presents a large-scale comparison between LLM-generated ideas and those produced by 100 NLP expert researchers, revealing that LLMs can produce ideas deemed more novel than human-generated ideas . . .
-

Current Trends:

- $I_{-3} \rightarrow I_{-2}$: The progression from I_{-3} to I_{-2} marks a significant shift from the application of neural models for molecular generation to the broader scope of automating scientific research using LLMs . . .
 - $I_{-2} \rightarrow I_{-1}$: The transition from I_{-2} to I_{-1} focuses on refining the autonomous induction capabilities of LLMs, specifically in generating novel and valid scientific hypotheses . . .
 - $I_{-1} \rightarrow I_0$: I_0 builds on the advancements made in I_{-1} by further extending the process of generating hypotheses to generating and refining research ideas autonomously . . .
 - $I_0 \rightarrow I_1$: The transition from I_0 to I_1 emphasizes the importance of empirical validation of LLMs in generating novel research ideas and highlights the potential of LLMs to contribute to ideation . . .
-

Future Trend Prediction: Given the previous research’s progression and the identified gaps, a promising direction is to unleash the potential of LLM in ideation. We can develop a multi-agent system that leverages evolutionary algorithms to enhance the diversity and novelty of LLM-generated research ideas . . .

Final Idea: *EvoResearchAgent: Enhancing Diversity and Novelty in Idea Generation with Evolution*

- **Motivation:** Using LLMs for idea generation has shown promising advancements. However, challenges persist, particularly concerning the diversity and novelty of LLM-generated ideas. Si et al. (2024) show that while LLMs can produce novel ideas, they often lack a broad range of perspectives and diversity. Additionally, Baek et al. (2024) have emphasized the need for a more systematic approach to improving the quality of generated ideas. To address these issues, we propose EvoResearchAgent, a multi-agent system that leverages evolutionary algorithms to enhance the diversity and novelty of generated ideas . . .
 - **Method:**
 - **Idea Initialize:** An LLM generates some initial ideas as the start point of the evolutionary process . . .
 - **Metrics:** Propose automatic metrics like topic diversity and novelty to evaluate the range of ideas . . .
 - **Evolution Integration:**
 1. **Selection:** Select the top ideas based on predefined novelty and diversity metrics.
 2. **Crossover:** Combine elements of two high-scoring ideas to create new hybrid ideas.
 3. **Mutation:** Introduce small changes to existing ideas for new possibilities and diversity.
 4. **Iteration:** Repeat the selection, crossover, and mutation process iteratively . . .
-

4.5 LENGTH OF CoI

To examine the impact of the CoI length on the quality of generated ideas, we constructed variants with differing maximum chain lengths. Furthermore, we also adopt the “- CoI” variant in Sec. 4.2 as a 0-length variant, which uses 5 retrieved papers but does not organize them in a chain structure. Figure 6 presents the idea arena results among these length variants. We observe a substantial improvement of idea-generation quality when we increase the length from 0 to 3. This indicates a clear developmental trend analysis is more pivotal than the quantity of related literature. Furthermore, the quality of generated ideas continues to improve as the length of the CoI increases. Longer CoIs offer more reliable and comprehensive insights into

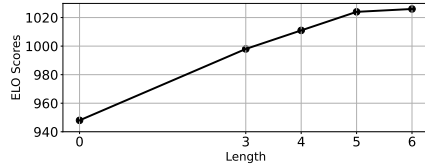


Figure 6: Length analysis of the CoI.